Linear Algebra

STAT 3160: Applied Multivariate Analysis

Weihua Zhou

January 18, 2022

Outline

- Vectors
- Matrices
- Useful Matrix Formulas
- Matrix Software (SAS/IML)

Vectors

Vectors: An array $\mathbf{x}$ of n real numbers is called a vector & written as

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \quad \text{or} \quad \mathbf{x}' = (x_1, x_2, \ldots, x_n)$$

- Vectors are denoted by lower case bold letters (i.e., when hand written, put a $\sim$ underneath the letter).
- Typically, vectors are columns.
- Transpose of a (column) vector is $\mathbf{x}'$ is a row vector.
- A vector consists of n scalars (numbers).
- A vector is a matrix with one column, $(n \times 1)$.

Vector as a Point

A Vector is a point in n-dimensional space represented by an arrow originating at zero.
All vectors have a *Length* and a *Direction*. e.g.,

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}.$$

Length of Vector

Let

$$\mathbf{x}' = (x_1, x_2, \ldots, x_n)$$

and the length of $\mathbf{x}$ is

$$L_x = \sqrt{x_1^2 + x_2^2 + \ldots + x_n^2} = \sqrt{\sum_{j=1}^{n} x_j^2}$$

Definition of Equal Vectors

Two vectors (matrices) are equal if they have the same number of rows, the same number of columns, and the corresponding entries in each are equal

$$\mathbf{a} = \mathbf{b} \quad \text{means} \quad a_{ij} = b_{ij} \quad \text{for all } i \ \& \ j$$

OR

Two vectors are equal (identical) if they both have the same length and direction.

Vector operation

- Suppose that c is some number (a scalar). For scalar multiplication, multiply each entry of $\mathbf{x}$ by c:

$$c\mathbf{x}' = (cx_1, cx_2, \ldots, cx_n)$$

- The sum of two vectors, each with the same number of elements (i.e., same size), element by element.

$$\mathbf{z} = \mathbf{x} + \mathbf{y} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} + \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} x_1 + y_1 \\ x_2 + y_2 \\ \vdots \\ x_n + y_n \end{pmatrix}$$

Geometry of Vector Addition

Form a parallelogram by sliding the origin of one vector to the end of the other.
The sum is the diagonal of the parallelogram. e.g.,

Distance

- Distance between Two Arbitrary Points: The "straight-line" distance between two arbitrary points $P = (x_1, x_2, \ldots, x_n)$ and $Q = (y_1, y_2, \ldots, y_n)$ is

$$d(P, Q) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + \ldots + (x_n - y_n)^2}$$

- Squared Distances: All points that are a constant (squared) distance, say c^2 from the origin $\mathbf{0}$ satisfy the equation

$$d^2(0, P) = x_1^2 + x_2^2 + \ldots + x_n^2 = c^2$$

  This is the equation for a sphere in n-dimensional space in (un-weighted) Euclidean space.

- Statistical Distance: Weight scores by a variable's standard deviation (more about these later). This yields an ellipsoid, which has important statistical connections.

Inner Product of Two Vectors in 2 dimensional space

Note

$$\cos(\theta) = \frac{x_1 y_1 + x_2 y_2}{\sqrt{x_1^2 + x_2^2}\sqrt{y_1^2 + y_2^2}}$$

The "inner product" of vectors $\mathbf{x}$ and $\mathbf{y}$ equals $(x_1 y_1 + x_2 y_2)$.

- Also called the "dot product".
- Denoted by $\mathbf{x}'\mathbf{y}$ or $\mathbf{y}'\mathbf{x}$ (order doesn't matter).
- $\mathbf{x}'\mathbf{y} = x_1 y_1 + x_2 y_2$
- $\mathbf{x}'\mathbf{x} = x_1^2 + x_2^2 = L_{\mathbf{x}}^2$
- $\mathbf{y}'\mathbf{y} = y_1^2 + y_2^2 = L_{\mathbf{y}}^2$
- $\cos(\theta) = 0$ when $\theta = 90^o \Leftrightarrow \mathbf{x}$ and $\mathbf{y}$ are perpendicular when $\mathbf{x}'\mathbf{y} = 0$.

Generalization to n dimensions

If $\mathbf{x}$ and $\mathbf{y}$ are $n \times 1$ vector, i.e.,

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \quad \text{and} \quad \mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$$

Then the inner product between $\mathbf{x}$ and $\mathbf{y}$ is

$$\mathbf{x}'\mathbf{y} = (x_1 y_1 + x_2 y_2 + \ldots + x_n y_n) = \text{a scalar}$$

$$\text{Length of } \mathbf{x} = L_{\mathbf{x}} = \sqrt{\mathbf{x}'\mathbf{x}}$$

$$\cos(\theta) = \frac{\mathbf{x}'\mathbf{y}}{L_{\mathbf{x}} L_{\mathbf{y}}} = \frac{\mathbf{x}'\mathbf{y}}{\sqrt{\mathbf{x}'\mathbf{x}}\sqrt{\mathbf{y}'\mathbf{y}}}$$

$\mathbf{x}'\mathbf{y} = 0$ if and only if $\mathbf{x}$ and $\mathbf{y}$ are perpendicular (orthogonal).

Matrices

- A Matrix is any rectangular array of real numbers. We'll use upper case bold Roman letters to represent them.
- An arbitrary matrix with n rows and p columns

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1p} \\ a_{21} & a_{22} & \dots & a_{2p} \\ \vdots & \vdots & \vdots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{np} \end{pmatrix}$$

- The first subscript indicates the row and second indicates a column.
- A matrix with n rows and p columns is said to have order $(n \times p)$.
- Another way to indicate a matrix (and it's order) is

$$\mathbf{A} = \{a_{ij}\}_{n \times p}$$

A Matrix is any rectangular array of real numbers. We'll use upper case bold Roman letters to represent them.

Matrix Operations & Definitions of Special ones

- *Transpose* operation is indicated by $'$ (e.g., $\mathbf{A}'$).

$$\mathbf{A}' = \mathbf{B} \Longleftrightarrow b_{ij} = a_{ji}$$

Example:

$$\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & 5 \\ 0 & 4 \\ 4 & 8 \end{pmatrix} \quad \text{and} \quad \mathbf{A}' =$$

Matrix Operations & Definitions of Special ones

(continued)

- *Equal matrices*:

$$\mathbf{A} = \mathbf{B} \text{ if and only if } a_{ij} = b_{ij} \text{ for all } i \text{ and } j$$

- *Scalar Multiplication*: multiply each element of **A** by the scalar c.

$$c\mathbf{A} = \begin{pmatrix} ca_{11} & ca_{12} & \dots & ca_{1p} \\ ca_{21} & ca_{22} & \dots & ca_{2p} \\ \vdots & \vdots & \vdots & \vdots \\ ca_{n1} & ca_{n2} & \dots & ca_{np} \end{pmatrix}$$

Matrix addition

- Addition is element-by-element.
- Two matrices must have the same order (referred to as conformal for addition).
- Example:

$$\mathbf{C} = \mathbf{A} + \mathbf{B} = \begin{pmatrix} 1 & 2 \\ 3 & 5 \\ 0 & 4 \\ 4 & 8 \end{pmatrix} + \begin{pmatrix} 1 & -2 \\ 3 & 1 \\ -3 & 4 \\ 4 & 8 \end{pmatrix}$$

Matrix addition

(continued)

- Matrix addition is commutative (i.e., order does not matter): $\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$.
- Matrix addition is associative: $(\mathbf{A} + \mathbf{B}) + \mathbf{C} = \mathbf{A} + (\mathbf{B} + \mathbf{C})$

Matrix Multiplication

- To be able to perform the matrix multiplication $\mathbf{C} = \mathbf{A} \times \mathbf{B}$, the number of columns in the matrix $\mathbf{A}$ must equal the number of rows in the matrix $\mathbf{B}$.
- The element in the ith row and jth column of the answer matrix $\mathbf{C}$, is the cross product sum of the ith row of the matrix $\mathbf{A}$ and the jth column of the matrix $\mathbf{B}$. This is done of all combinations of rows of $\mathbf{A}$ and columns of $\mathbf{B}$.
- Example:

$$\begin{pmatrix} 1 & 2 \\ 2 & 3 \\ 4 & 3 \end{pmatrix} \times \begin{pmatrix} 4 & 2 \\ 1 & 3 \end{pmatrix} =$$

More Properties of Matrix Operations

- Matrix multiplication is associative:

$$\mathbf{ABC} = (\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC})$$

- Matrix multiplication is Distributive
  1. Left distributive: $\mathbf{AB} + \mathbf{AC} = \mathbf{A}(\mathbf{B} + \mathbf{C})$
  2. Right distributive: $\mathbf{AC} + \mathbf{BC} = (\mathbf{A} + \mathbf{B})\mathbf{C}$
- Properties of transposes:
  1. $(\mathbf{AB})' = \mathbf{B}'\mathbf{A}'$ and $(\mathbf{ABC})' = \mathbf{C}'\mathbf{B}'\mathbf{A}'$.
  2. $(\mathbf{A}')' = \mathbf{A}$
  3. $(\mathbf{A} + \mathbf{B})' = \mathbf{A}' + \mathbf{B}'$

Some Special Matrices

- *Symmetric matrix*. A Symmetric Matrix is a square matrix where the rows equal the columns

$$\mathbf{A} = \mathbf{A}', \quad i.e., \quad a_{ij} = a_{ji}$$

  1. Important examples of symmetric matrices in multivariate statistics include the variance-covariance matrix and the correlation matrix.
  2. Example:

$$\mathbf{A} = \mathbf{A}' = \begin{pmatrix} 4 & 2 & 1 \\ 2 & 3 & 0 \\ 1 & 0 & 5 \end{pmatrix}$$

- *Null Matrix*. A Null Matrix has all elements equal to 0.
- *Trace of a Square Matrix*. The sum of main diagonal entries of a square matrix is called the trace, e.g.,

$$\text{trace}(\mathbf{A}) = 4 + 3 + 5 = 12.$$

Some Properties of Matrix Trace

Let $\mathbf{A} = \{a_{ij}\}$ be a $p \times p$ square matrix.

- $\text{tr}(c\mathbf{A}) = c\text{tr}(\mathbf{A})$
- $\text{tr}(\mathbf{A} \pm \mathbf{B}) = \text{tr}(\mathbf{A}) \pm \text{tr}(\mathbf{A})$
- $\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA})$
- $\text{tr}(\mathbf{B}^{-1}\mathbf{AB}) = \text{tr}(\mathbf{A})$
- $\text{tr}(\mathbf{AA}') = \sum_{i=1}^{p} \sum_{j=1}^{p} a_{ij}^2$.

Some Special Matrices(continued)

- *Diagonal Matrix*. A Diagonal Matrix is a square matrix where the only non-zero elements are on the diagonal, e.g.,

$$\mathbf{D}_{(p\times p)} = \begin{pmatrix} d_{11} & 0 & \dots & 0 \\ 0 & d_{22} & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \dots & d_{pp} \end{pmatrix}$$

- *Identity Matrix*. The Identity Matrix is a diagonal matrix with all entries equal to 1, e.g.,

$$\mathbf{I}_p = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}$$

$\mathbf{I}_p$ is call the identity matrix because

$$\mathbf{I_p A}_{(p\times p)} = \mathbf{A}_{(p\times p)}\mathbf{I}_p = \mathbf{A}_{(p\times p)}$$

Determinant of a Matrix

The determinant of a square matrix, $\mathbf{A}_{(p\times p)}$, is a scalar that is,associated with that matrix. Determinants are denoted by $|\mathbf{A}|$ or $\det(\mathbf{A})$.

How to compute the determinant of

- 1×1 matrix:

$$|a| = a$$

- 2×2 matrix:

$$\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \longrightarrow |\mathbf{A}| = ad - bc$$

- 3×3 matrix:

$$\mathbf{A} = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \longrightarrow |\mathbf{A}| = aei + dhc + gfb - (ceg + fha + idb)$$

Determinant of a $(p \times p)$ Matrix

For $p > 3$, this requires the use a recursive process, but first need some definitions:

- Let $\mathbf{A}_{uv}$ be the $(p-1) \times (p-1)$ submatrix of $\mathbf{A}$ constructed by deleting the uth row and vth column of $\mathbf{A}$.
- Minor entry of a_{uv} is the $\det(\mathbf{A}_{uv})$.
- The signed minor of $(-1)^{u+v}\det(\mathbf{A}_{uv})$ is the cofactor of a_{uv} .

The recursive algorithm:

1. Chose some row (or column) either arbitrarily or for convenience.
2. Find the cofactors for the entries in it.
3. The cofactors would then be weighted by the relevant entries and summed. i.e.,

$$\det(\mathbf{A}) = \sum_{u=1}^{p} a_{uv}(-1)^{u+v}\det(\mathbf{A}_{uv})$$

Example: Determinant of a (3×3) matrix

$$\mathbf{A} = \begin{pmatrix} 4 & 3 & 2 \\ 2 & 5 & 1 \\ 1 & 3 & 2 \end{pmatrix}$$

More About Determinants

- The determinant of a diagonal matrix is equal to the product of the diagonal elements:

$$\mathbf{A}_{(p\times p)} = \begin{pmatrix} a_{11} & 0 & \dots & 0 \\ 0 & a_{22} & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \dots & a_{pp} \end{pmatrix} \longrightarrow \det(\mathbf{A}) = \prod_{j=1}^{p} a_{jj}$$

- The determinant of a upper- (or lower-) triangular matrix equals the product of the diagonal elements:

$$\mathbf{A}_{(p\times p)} = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1p} \\ 0 & a_{22} & \dots & a_{2p1} \\ \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \dots & a_{pp} \end{pmatrix} \longrightarrow \det(\mathbf{A}) = \prod_{j=1}^{p} a_{jj}$$

Properties of Determinants

- If one row (or column) of $\mathbf{A}$ is multiplied by a constant c, the new determinant is $c\det(\mathbf{A})$.
- If two rows or two columns of a matrix are interchanged, the sign of the determinant is changed.
- If two rows or two columns of a matrix are equal, then the determinant is zero.
- The determinant is unchanged by adding a multiple of some row to another row (the same is true for columns).
- A zero row or column implies a zero determinant.
- $\det(\mathbf{AB}) = \det(\mathbf{A})\det(\mathbf{B})$

Inverse of a Matrix

An inverse (in the traditional sense) can be found only for square matrices.
Defintion: The inverse of a square matrix $\mathbf{A}$ is the matrix $\mathbf{A}^{-1}$ such that $\mathbf{A}^{-1}\mathbf{A} = \mathbf{A}\mathbf{A}^{-1} = \mathbf{I}$.

- The inverse exists if and only if the p columns (or rows) of $\mathbf{A}_{(p\times p)}$ are linearly independent. i.e.,

$$c_1\mathbf{a}_1 + c_2\mathbf{a}_2 + \ldots + c_p\mathbf{a}_p = 0$$

  where $c_1 = c_2 = \ldots = c_k = 0$.
- If $\mathbf{A}^{-1}$ exists, $\mathbf{A}$ is nonsingular. If $\mathbf{A}^{-1}$ does not exists, then $\mathbf{A}$ is singular.

Inverse of a Matrix

- Computing inverse for 2×2 is easy:

$$\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \longrightarrow \mathbf{A}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

  Example:

$$\mathbf{A} = \begin{pmatrix} 10 & 6 \\ 8 & 5 \end{pmatrix}$$

- Inverse of a Diagonal Matrix:

$$\mathbf{A} = \begin{pmatrix} a_{11} & 0 & \dots & 0 \\ 0 & a_{22} & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \dots & a_{pp} \end{pmatrix} \longrightarrow \mathbf{A}^{-1} = \begin{pmatrix} a_{11}^{-1} & 0 & \dots & 0 \\ 0 & a_{22}^{-1} & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \dots & a_{pp}^{-1} \end{pmatrix}$$

- The calculation of an inverse for large matrices is a laborious process that we'll leave to the computer.

Quadratic Forms

- Suppose $\mathbf{A}_{n\times n}$ is symmetric and let $\mathbf{x}' = (x_1, \ldots, x_n)$. A quadratic form is defined by

$$\mathbf{x}'\mathbf{A}\mathbf{x} = \sum_{j=1}^{n}\sum_{k=1}^{n} a_{jk}x_jx_k$$

- Why are quadratic forms important? One example,. . . What does this equal?

$$(x_1, x_2, \ldots, x_n)\begin{pmatrix} n/(1-n) & -1/n & \ldots & -1/n \\ -1/n & n/(1-n) & \ldots & -1/n \\ \vdots & \vdots & \vdots & \vdots \\ -1/n & -1/n & \ldots & n/(1-n) \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$$

$= \sum_{j=1}^{n}(x_j - \bar{x})^2$

More about Quadratic Forms

- A symmetric matrix $\mathbf{A}$ (and associated quadratic form) is
  1. positive definite (p.d.) if $\mathbf{x}'\mathbf{A}\mathbf{x} > 0 \quad \text{for all} \quad \mathbf{x} \neq \mathbf{0}$.
  2. positive semi-definite (p.d.) if $\mathbf{x}'\mathbf{A}\mathbf{x} \geq 0 \quad \text{for all} \quad \mathbf{x} \neq \mathbf{0}$.
- Negative definite, negative semi-definite, and indefinite forms exist as well; however, correlation and covariance matrices
  1. Are at least positive semi-definite.
  2. Satisfies the stronger condition of being positive definite if the vectors of the variables on which the correlation and covariance matrices is based, are linearly independent.
- Matrices that are positive definite have inverses (i.e., nonsingular).

First Pass at Eigenvalues and Eigenvectors

- A square matrix $\mathbf{A}_{p\times p}$ has an eigenvalue λ (a scalar) and a corresponding eigenvector $\mathbf{x}(\neq 0)$ if

$$\mathbf{A}\mathbf{x} = \lambda\mathbf{x} \quad \text{or equivalently} \quad (\mathbf{A} - \lambda\mathbf{I})\mathbf{x} = 0$$

- Note

$$\begin{aligned} \mathbf{A}\mathbf{x} &= \lambda\mathbf{x} \\ \mathbf{A}\mathbf{x} - \lambda\mathbf{x} &= 0 \\ \mathbf{A}\mathbf{x} - \lambda\mathbf{I}\mathbf{x} &= 0 \\ (\mathbf{A}\mathbf{x} - \lambda\mathbf{I})\mathbf{x} &= 0 \end{aligned}$$

- Usually we divide $\mathbf{x}$ by it's length $L_{\mathbf{x}}$ so that we get a normalized eigenvector; eigenvectors with length $= 1$,

$$\mathbf{e} = \frac{\mathbf{x}}{L_{\mathbf{x}}} \longrightarrow \mathbf{e}'\mathbf{e} = 1$$

spectral decomposition

- Let $\mathbf{A}$ be $p \times p$ square symmetric matrix. Then $\mathbf{A}$ has p pairs of eigenvalues and eigenvectors namely,

$$\lambda_1, \mathbf{e}_1, \lambda_2, \mathbf{e}_2, \ldots, \lambda_p, \mathbf{e}_p,$$

  The eigenvectors can be chosen to satisfy $\mathbf{e}_1'\mathbf{e}_1 = \ldots = \mathbf{e}_p'\mathbf{e}_p = 1$ and be mutually orthogonal, that is, $\mathbf{e}_i'\mathbf{e}_j = 0$. The eigenvectors are unique unless two or more eigenvalues are equal.

- Furthermore, one has the following spectral decomposition:

$$\begin{aligned} \mathbf{A} &= \lambda_1\mathbf{e}_1\mathbf{e}_1' + \lambda_2\mathbf{e}_2\mathbf{e}_2' + \ldots + \lambda_p\mathbf{e}_p\mathbf{e}_p' \\ &= \sum_{j=1}^{p} \lambda_j\mathbf{e}_j\mathbf{e}_j' \end{aligned}$$

Orthogonal Matrices

- $\mathbf{A}_{p\times p}$ is an orthogonal matrix if for all pairs of columns (rows)

$$\mathbf{a}_i'\mathbf{a}_k = \begin{cases} 0 & \text{for } i \neq k \\ 1 & \text{for } i = k \end{cases}$$

- If $\mathbf{A}$ is orthogonal, then $\mathbf{A}' = \mathbf{A}^{-1}$; that is, $\mathbf{A}\mathbf{A}^{-1} = \mathbf{I} = \mathbf{A}\mathbf{A}'$ and $\mathbf{A}^{-1}\mathbf{A} = \mathbf{I} = \mathbf{A}'\mathbf{A}$.
- If $\mathbf{A}$ is orthogonal, then $\det(\mathbf{A}) = \pm 1$.
- If $\mathbf{A}$ and $\mathbf{B}$ are both orthogonal, the $\mathbf{AB}$ is also orthogonal.
- Lengths of vectors don't change under orthogonal transformations; that is, let $\mathbf{y} = \mathbf{A}\mathbf{x}$; then

$$L_{\mathbf{y}}^2 = \mathbf{y}'\mathbf{y} = (\mathbf{A}\mathbf{x})'(\mathbf{A}\mathbf{x}) = \mathbf{x}'\mathbf{A}'\mathbf{A}\mathbf{x} = \mathbf{x}'\mathbf{I}\mathbf{x} = \mathbf{x}'\mathbf{x} = L_{\mathbf{x}}^2$$

A Square-root Matrix

The spectral decomposition allows us to express the inverse of a square matrix in terms of its eigenvalues and eigenvectors, and this leads to a useful *square-root matrix*. Let $\mathbf{A}$ be a $p \times p$ positive definite matrix with spectral decomposition $\mathbf{A} = \sum_{j=1}^{p} \lambda_j \mathbf{e}_j \mathbf{e}_j'$.

- $$\mathbf{A} = \sum_{j=1}^{p} \lambda_j \mathbf{e}_j \mathbf{e}_j' = \mathbf{P}\boldsymbol{\Lambda}\mathbf{P}'$$

  where $\boldsymbol{\Lambda}$ is a $p \times p$ diagonal matrix

  $$\boldsymbol{\Lambda} = \begin{pmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \dots & \lambda_p \end{pmatrix},$$

  and $\mathbf{P} = [\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_p]$ with

  $$\mathbf{P}\mathbf{P}' = \mathbf{P}'\mathbf{P} = \mathbf{I}.$$

A Square-root Matrix (continued)

- $$\mathbf{A}^{-1} = \mathbf{P}\boldsymbol{\Lambda}^{-1}\mathbf{P}' = \sum_{j=1}^{p} \frac{1}{\lambda_j}\mathbf{e}_j\mathbf{e}_j'.$$

- Define the *square root* of $\mathbf{A}$ by $\mathbf{A}^{1/2}$ where

$$\mathbf{A}^{1/2} = \sum_{j=1}^{p} \sqrt{\lambda_j}\mathbf{e}_j\mathbf{e}_j' = \mathbf{P}\boldsymbol{\Lambda}^{1/2}\mathbf{P}'.$$

1. $(\mathbf{A}^{1/2})' = \mathbf{A}^{1/2}$.
2. $\mathbf{A}^{1/2}\mathbf{A}^{1/2} = \mathbf{A}$.
3. $$(\mathbf{A}^{1/2})^{-1} = \sum_{j=1}^{p} \frac{1}{\sqrt{\lambda_j}}\mathbf{e}_j\mathbf{e}_j' = \mathbf{P}\boldsymbol{\Lambda}^{-1/2}\mathbf{P}',$$

   where $\boldsymbol{\Lambda}^{-1/2}$ is a diagonal matrix with $1/\sqrt{\lambda_j}$ as the jth diagonal element.
4. $\mathbf{A}^{1/2}\mathbf{A}^{-1/2} = \mathbf{A}^{-1/2}\mathbf{A}^{1/2} = \mathbf{I}$, and $\mathbf{A}^{-1/2}\mathbf{A}^{-1/2} = \mathbf{A}^{-1}$, where $\mathbf{A}^{-1/2} = (\mathbf{A}^{1/2})^{-1}$.

Some properties related to eigen-system

Given a square matrix $\mathbf{A}_{p\times p}$ with eigenvalues $\lambda_1 \geq \lambda_2 \geq \ldots \geq \lambda_p$.

- Trace:

$$trace(\mathbf{A}) = \sum_{j=1}^{p} \lambda_j = \lambda_1 + \lambda_2 + \ldots + \lambda_p;$$

- Determinant:

$$det(\mathbf{A}) = |\mathbf{A}| = \prod \lambda_{\mathbf{j}} = \lambda_{\mathbf{1}}\lambda_{\mathbf{2}} \ldots \lambda_{\mathbf{p}}$$

- Furthermore, if $\mathbf{A}$ is a positive definite matrix. Then

$$\max_{\mathbf{x}\neq\mathbf{0}} \frac{\mathbf{x}'\mathbf{A}\mathbf{x}}{\mathbf{x}'\mathbf{x}} = \lambda_1 \quad (\text{attained when } \ \mathbf{x} = \mathbf{e_1})$$

$$\min_{\mathbf{x}\neq\mathbf{0}} \frac{\mathbf{x}'\mathbf{A}\mathbf{x}}{\mathbf{x}'\mathbf{x}} = \lambda_p \quad (\text{attained when } \ \mathbf{x} = \mathbf{e_p})$$